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August 30, 1949

FEDERAL SPECIFICATION

STENCIL, DUPLICATING MACHINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers duplicating stencil paper for use with mimeograph machines (see 6.1).

1.2 Classification.

1.2.1 Classes. The stencil paper covered by this specification shall be of the following classes, as specified (see 6.2).

Class 1 - With transparent, top film.
Class 2 - Without transparent, top film.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-C-621 - Correction Compound; Stencil.
TT-I-564 - Ink, Stencil, Duplicating Machine, Black (For Use on Single-Cylinder, Internally Inked, Stencil-Duplicating Machines).
UU-P-388 - Paper, Mimeograph.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Test Method Std. No. 101 - Preservation, Packaging and Packing Materials.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical Association of the Pulp and Paper Industry (TAPPI) Standards:

T 402 - Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets and Related Products.
T 404 - Tensile Breaking Strength of Paper and Paperboard.
T 410 - Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 456 - Wet Tensile Breaking Strength of Paper and Paperboard.
T 457 - Stretch of Paper and Paperboard.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Stencil paper. The stencil paper shall be a suitable, fibrous tissue, treated with wet strength resins and coated with a blue (unless otherwise specified), stencil-forming material.

3.1.1.1 Odor. The stencil paper shall have no discernible, disagreeable odor.

3.1.2 Backing sheet. The backing sheet shall be made from fibrous, cellulosic material.

3.1.3 Physical requirements. The stencil paper and the backing sheet shall conform to the physical requirements as specified in Table I.

TABLE I. Physical requirements

Characteristic	Requirements
Stencil	
Dry tensile, not less than, both directions	2.8 pounds/inch width
Elongation, not more than	5 percent
Backing sheet	
Basis weight, 25x38-500, not less than	60 pounds

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3.1.4 Cushioning sheet. The cushioning sheet shall be made of fibrous, cellulosic material, and may be coated or impregnated at the option of the supplier. This sheet shall provide a resilient cushion between the stencil paper and the backing sheet.

3.1.5 Top film. The top film for the stencils shall be a flexible, plastic film, selected at the option of the supplier. When used in place over the stencil paper, the film shall permit the cutting of the stencil and shall be transparent to the extent that the stenciled characters are clearly visible.

3.2 Sizes. Unless otherwise specified, the stencil paper size, as expressed by the overall dimensions of the backing sheet, shall be 8-1/2 by 10 inches or 8-1/2 by 14-1/4 inches, as specified (see 6.2). A tolerance of ± 1/16 inch in each direction of the paper will be permitted.

3.3 Construction. Each sheet of stencil paper shall be secured to a backing sheet 1-3/4 ± 1/8 inch from the upper edge of the backing sheet. The stencil paper shall be the same width as the backing sheet with a ± 1/16 inch tolerance. The lower edge of the stencil paper shall be no longer and not less than 3/16 inch short of the lower edge of the backing sheet. The backing sheet shall be perforated 2-1/8 ± 1/8 inches from the upper edge across its width, so that when the body of the backing sheet is detached there will remain a paper stub to which the stencil paper is attached. The stencil paper shall remain firmly adhered to the stub during the duplicating process. The top of the stub shall contain four holes, suitably spaced, for holding the stub in a duplicating machine (see 6.1.1). Class 1 stencils only shall have a top film adhered to the stub portion of the backing sheet, between the upper edge of the backing sheet and the stencil paper. The film shall be the width of the stencil paper and shall extend at least 1/4 inch beyond the lowest horizontal guideline. The film shall be so adhered as to remain in place when cutting the stencil but shall be easily removed without fiber displacement when the stencil is to be mounted on the duplicating machine. The backing sheet shall have a notch on the lower end, so as to facilitate separating of the stencil from the backing sheet.

3.4 Number and size of cushioning sheets. Not less than 8 sheets of coated or impregnated cushioning sheets or 12 sheets of non-coated or non-impregnated cushioning sheets per 24 stencils shall be furnished (see 5.1). Cushioning sheets shall be as wide as the stencil paper and as long as the available typing space when inserted between the stencil paper and the backing sheet.

3.5 Guidelines and scale numbers. Stencil paper shall have markings to indicate horizontal spacing, margins and scale numbers. The markings shall be imprinted in white (unless otherwise specified) on the stencil paper. Marginal markings shall be located 11/16 ± 1/16 inch from and parallel to the right and left edges of the stencil paper. Scale numbers, corresponding to line spaces on a typewriter, shall be indicated in the two margins and shall be numbered consecutively from top to bottom. For 8-1/2 by 10 inch stencil paper, scale numbers shall be 1 to 84 inclusive; for 8-1/2 by 14-1/4 inch stencil paper, scale numbers shall be 1 to 62 inclusive. The lengthwise space available for cutting the stencil shall be outlined by horizontal markings, perpendicular to the margin markings, located one line space above scale number 1 and at scale numbers 62 and 84 as applicable. Additional guide markings may be added at the option of the supplier.

3.6 Performance.

3.6.1 Duplicating quality. The stencils shall be capable of producing not less than 5000 copies without distortion or other failures that impair legibility. The cuts in the stencil shall be readily penetrated by the ink to give clear, distinct copies. All copies produced shall be legible, clear and uniform in depth of the color when tested as specified in 4.3.2.

3.6.2 Legibility, cut-outs and type filling. The stencil paper finish shall not give off a glare that will prevent the easy reading of the typed characters. The stencil paper shall show no evidence of cut-outs and the last impression shall be as clear and distinct as the tenth impression with respect to type filling when tested as specified in 4.3.1.

3.6.3 Permanence. The stencil paper shall retain its serviceability under normal storage conditions for not less than 12 months from time of delivery and shall withstand the tests for accelerated aging as specified in 4.3.3.1 and 4.3.3.2.

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3.6.4 Correction properties. There shall be no noticeable difference in the quality of duplication by stencils before and after correction compound has been applied when tested as specified in 4.3.2.

3.6.5 Use of stylus. The stencil paper shall show complete and uniform displacement of the coating without tearing or pulling the paper and without requiring a protecting sheet on top of the stencil when tested with a stylus as specified in 4.3.2.

3.7 Instruction sheet. Detailed instructions for the preparation of stencils shall be furnished with each package of stencils (see 6.2).

3.8 Greaseproof packages. Twenty-four complete stencil assemblies as described in 5.1 shall be packaged in an oil and greaseproof package. There shall be no staining, grease or oil spots on the white paper when tested according to Procedure B of Method 3017 in Fed. Test Method Std. No. 101.

3.9 Workmanship. The stencils shall be clean, free from holes, cuts, tears or wrinkles. All parts of the stencil assembly, stencil paper, backing sheet, cushioning sheet and top film (as applicable), shall be neatly trimmed and have smooth and clean cut edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Sampling for examination and tests shall be performed in accordance with MIL-STD-105, except where otherwise noted. For purposes of sampling, a lot shall consist of all stencils of one class submitted for inspection at one time.

4.1.2 Inspection of components and materials. In accordance with 4.1 the supplier is responsible for insuring that components and materials were manufactured, examined, and tested in accordance with this specification or referenced subsidiary specifications and standards.

4.2 Inspection of the end item.

4.2.1 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.2.1.5. Random samples shall be drawn from each lot for examination of visual, dimensional and preparation for delivery requirements. The lot size shall be expressed in units of stencil assemblies (including backing sheet and top film as applicable) for examinations in 4.2.1.3 and 4.2.1.4, in units of shipping containers for examination in 4.2.1.1 and in units of packages for examination in 4.2.1.2.

4.2.1.1 Examination of preparation for delivery defects. An examination shall be made to determine whether markings, materials and workmanship comply with the requirements of section 5. The sample unit shall be one fully packed, shipping container. Defects shall be scored as listed below.

Examine	Defects
Markings	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence or method of application.
Materials	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose or inadequate sealing, strapping or stapling.
	Distorted or punctured container.
Contents	Number of intermediate packages per shipping container not as specified.

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4.2.1.2 Examination for defects in count and contents of unit packages. The sample unit shall be one unit package and shall be randomly selected from each of the shipping containers selected in 4.2.1.1. Defects shall be scored as listed below.

Examine	Defects
Count	Less than specified.
Cushioning sheets	Less than eight coated or impregnated sheets or 12 uncoated sheets.
Instructions	Missing or not as specified.

4.2.1.3 Examination of the end item for visual defects. The sample unit for this examination shall be one complete stencil assembly with backing sheet and, as applicable, top film. At least one sample unit shall be taken from each shipping container selected for examination in 4.2.1.1. Defects shall be scored as listed below.

Examine	Defects
Form	Not class specified.
Construction	Backing sheet missing, not perforated or not perforated full width. Stub end of backing sheet contains more or less than four holes; holes torn through to edge or to adjacent hole.
Class 1, only	Top film missing; not adhered to stub as specified; not easily removed.
Appearance	Guidelines, not as specified or missing. Scale numbers, not as specified for sheet size or missing. Top film not flat over stencil portion; wrinkled, creased or ridged.
Workmanship	Not clean, presence of holes, tears, cuts or wrinkles. Edges not clean cut.

4.2.1.4 Examination for dimensional defects. The sample unit for this examination shall be one stencil assembly and one cushion sheet. At least one sample unit shall be taken from each shipping container selected for examination in 4.2.1.1. Defects shall be scored as listed below.

Examine	Defects
Backing sheet	Varies from specified size by more than ± 1/16 inch. Perforations located less than 2 inches or more than 2-1/4 inches from upper edge.
Stencil paper	Width not within ± 1/16 inch of the backing sheet. Not adhered to backing sheet 1-3/4 ± 1/8 inch from upper edge. More than 3/16 inch short of the lower edge of backing sheet or longer than the backing sheet.
Top film, as applicable	Width more or less than stencil paper. Does not extend beyond bottom guideline by 1/4 inch or more.
Cushioning sheets	Width more or less than stencil paper. Length less than space available for stencil cutting for the applicable size stencil.
Guidelines	Marginal markings located less than 5/8 inch or more than 3/4 inch in from right and left hand edges. Horizontal guidelines not located as specified.

4.2.1.5 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels for determining the sample size and acceptable quality levels (AQLs) expressed as defects per 100 stencils for examination in 4.2.1.3 and 4.2.1.4, as defects per 100 intermediate packages in 4.2.1.2 and as defects per 100 shipping containers in 4.2.1.1 shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.1.1	S-2	2.5
4.2.1.2	S-1	2.5
4.2.1.3	S-4	1.5
4.2.1.4	S-2	1.5

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4.2.2 Testing of the end item. The end item shall be tested for the applicable characteristics in table II. The methods of testing specified in table II shall be followed. The lot size shall be expressed in units of stencils. The test sample unit shall consist of 5 stencil sheet assemblies, including backing sheet, top film, as applicable, stencil and one cushion sheet. The inspection level shall be S-1. Not more than one sample unit shall be drawn from any one package. Test results shall include all values upon which the results are based and there shall be no failure to meet the sample unit or lot average requirements, as applicable. All samples for testing, except for the performance testing (see table II), shall be conditioned in accordance with TAPPI T 402.

TABLE II. Instructions for testing

Characteristic	Specification reference		Requirements applicable to		Number determinations per unit	Results reported as	
	Rqmt	Test method	Sample Unit	Lot avg.		Pass or fail	Numerically to nearest
Atmospheric conditions	4.3.2	T402 1/	All testing except performance tests				
Stencil paper							
Dry tensile	3.1.3	T404 1/		X	10 each direction		0.1 pound
Elongation	3.1.3	T457 1/		X	10 each direction		1 percent
Backing sheet							
Basis weight	3.1.3	T410 1/		X	2		0.1 pound
Performance							
Duplicating quality	3.7.1	4.3.2	X		1	X	
Legibility, cut-outs, type filling	3.7.2	4.3.1	X		1	X	
Permanence	3.7.3	4.3.3	X		1	X	
Correction properties	3.7.4	4.3.2	X		1	X	
Stylus tracing	3.7.5	4.3.2	X		1	X	
Stencil package	3.9	3017 2/ Procedure B	X		1	X	

1/ Refers to TAPPI Test Methods.
2/ Fed. Std. No. 101.

4.3 Test methods.

4.3.1 Legibility, cut-outs, type filling. Use an electric typewriter with clean, elite type and a soft, platen roll designed for cutting stencils on the machine. Insert the stencil paper with top film, as applicable, and cushioning sheet, in accordance with the manufacturer's instructions, and beginning at the top of the page cut the stencil paper with the entire alphabet of both capital and small letters, repeating each letter not less than 10 times. Advance the stencil one or more spaces and type a block of small letter "e's" (approximately 300), and examine the stencil paper for visibility of stenciled characters and cut-outs when viewed through the top film, as applicable. Examine the loop of the last "e" in the block of small"e's" for evidence of typing filling (loop not distinct; type filled with stencil paper material). Note also if any glare is given off by the stencil sheet background. If the stencil sheet gives off glare causing the stencil to be difficult to read and contains cut-outs and indistinct letters, the stencil paper shall be considered as failing to meet the specification requirements.

4.3.2 Duplicating quality, correction properties, stylus tracing. Take correction compound complying with TT-C-621 and block out a typed line of letters cut in test specified in 4.3.1. Using same typewriter under the conditions specified in 4.3.1, insert stencil paper and type over the blocked-out line. Remove the stencil, clean the platen roll by wrapping a sheet of blotting paper around it and rolling it back and forth several times. Also release the feed rolls. In the lower unused portion of the stencil paper trace various circular and rectangular figures, using a minimum looped stylus. Place the cut stencil in a mimeograph machine. Using type II mimeograph paper

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as specified in UU-P-388 and ink in accordance with type I of TT-I-564 run off at least 5000 copies. Retain the first 50 copies, the 100th and 500th, and each additional 500th copy up to and including the 5000th copy. Examine retained copies for correction properties, tracing ability, and printing quality. Failure of the stencil paper to take corrections properly, failure to work satisfactorily with a stylus, or failure to show satisfactory duplicating quality will be considered as noncompliance with the requirements of this specification.

4.3.3 Permanence.

4.3.3.1 Accelerated aging. Take 20 alternate strips each of the stencil paper and backing, 1/2 x 4 inches, and place them between a pair of wooden blocks 1 x 4 x 1/2 inch which have been previously dried at 105° ± 2°C. Place each assembly in an oven in which the temperature is maintained uniform throughout, with a 2-pound weight on each pair of blocks, and heat for 3 hours at 105°C.(± 2°C.). At the end of the 3-hour heating period, the paper shall show no tendency to stick to the backing sheet, no evidence of brittleness, and when compared with the original unheated portions of the sample, there shall be no appreciable change of color or appreciable change in the backing sheet.

4.3.3.2 Duplicating quality of accelerated-aged stencil paper. Take one sheet of stencil paper from each package, and cut it down the middle making two pieces approximately 4-1/4 inches wide and 12 inches long overall. Save one half of each sheet as a control. Take the other half, remove the backing sheet and plastic film, as applicable, and place the stencil sheet in an oven in which the temperature is maintained uniform throughout and heat at 105° ± 2°C. for 1 hour. Then mount the two complementary halves of each stencil sheet on one backing sheet and cut the stencil as described in 4.3.2 on each half. Place the two halves as a unit on a mimeograph machine and run at a speed of 5000 impressions per hour until at least 1000 copies are prepared. There shall be no difference of duplicating properties of the heated and unheated portions.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Twenty-four complete stencils described in 3.3, information instructions described in 3.8, and either 8 coated or impregnated or 12 non-coated or nonimpregnated cushion sheets described in 3.4 shall be packaged in box, wrapper or an envelope which shall meet the test requirements specified in 4.2.2.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Forty packages of stencils specified in 5.1 shall be packed in a box conforming to PPP-B-636, class weather resistant.

5.2.2 Level B. Forty packages of stencils specified in 5.1 shall be packed in a box conforming to PPP-B-636, class domestic.

5.2.3 Level C. Stencils in quantities as specified (see 6.2), packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special markings required in the procurement documents, marking of the interior packages and exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable (see 6.2).

6. NOTES

6.1 Intended use. The stencils covered by this specification are intended for use as masters for reproducing on mimeograph duplicating machine.

6.1.1 Duplicating machine. For use with, but not limited to the following mimeograph machine models: A.B. Dick - 76, 77A, 90, 77B, 78, 91, 93, 96, 100, 400 series; 530, 550; Heger - C, D, E, 24; Ideal - G, F, E; Marr - B, B-2, E, G, G-2; Niagara - 150-E, 110-E; Print-O-Matic - 5-A, 6-A, 5-E, 5-B; BDC Rex-rotary - M-4EAU, M-4HAU, M-4E, M-2; Rex-rotary - D-28-DEAU, D280HAU, D280H, D280ESA; Speed-O-Print (Liberator Series) - 50, 75, 100, 200, 300; Tempo - 79; Gestetner - 300, 310, 320, 330, 360, 466,460,420,410,400, 145.

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6.1.2 The use of film topped stencils (class 1) reduces cutout of letters when the typewriter type is too sharp, or where a hard platen, accompanied by a heavy stroke is encountered. The copy quality resulting when using film topped stencils will be bolder and blacker but may also result in offsetting (image - blurring) and show-through due to excessive ink deposition on the copy paper.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Quantities required (see 5.2.3).
(e) Marking document required (see 5.3).

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